

Question 1

Based on the Mock Logical ERD - Retrieve a list of device makes and models with the names and position of staf to whom they have been allocated. Also add the staff who have not been allocated a device.

A SELECT De.deviceMake, De.deviceModel, Em.IName, Em.position
FROM Device De LEFT OUTER JOIN Emp Em
ON Em.deviceId = De.deviceId;

B SELECT De.deviceMake, De.deviceModel, Em.IName, Em.position
FROM Device De RIGHT OUTER JOIN Emp Em
ON De.empId = Em.empId;

C SELECT De.deviceMake, De.deviceModel, Em.IName, Em.position
FROM Device De LEFT OUTER JOIN Emp Em
ON De.empId = Em.empId;

D SELECT De.deviceMake, De.deviceModel, Em.IName, Em.position
FROM Emp Em RIGHT OUTER JOIN Device De
ON Em.empId = De.empId;

Question 2

Based on the XML document uni_marks.xml and using XPath - Return the surnames of the students that take either the module MOD101 or MOD103.

- ☐ **A** // module [@moduleCode = "MOD101" and @moduleCode = "MOD103"] // sname
- ☐ **B** // module [@moduleCode = "MOD101" | @moduleCode="MOD103"] // sname
- ☒ **C** // module [@moduleCode = "MOD101" or @moduleCode = "MOD103"] // sname
- ☐ **D** // module [@moduleCode = "MOD101"] or // module [@moduleCode = "MOD103"] // sname

Question 3

Based on the XML document uni_marks.xml and using XPath - Return the marks scored by student ID 21033 in a TEST for any components of the assessment.

- ☐ A // student [@studentId = 21033] / mark1 [@type = "test"]
and // student [@studentId = 21033] / mark2 [@type = "test"]
- ☒ B // student [@studentId = 21033] / mark1 [@type = "test"]
| // student [@studentId = 21033] / mark2 [@type = "test"]
- ☐ C // student [@studentId = 21033] / mark1 [@type = "test"]
or // student [@studentId = 21033] / mark2 [@type = "test"]
- ☐ D // student [@studentId = 21033] / [@type = "test"] [mark1 or mark2]

Question 4

Based on the Mock Logical ERD - Retrieve a list of surnames, hire dates, department names and device makes for those staff in these departments who have been allocated a device that was purchased from the 16 February 2021 onwards.

- A

```
SELECT E.IName, E.hireDate, D.dName, De.deviceMake, De.devicePurchDate
FROM Dept D
JOIN Emp E ON D.deptNo = E.deptNo
JOIN Device De ON De.emplId = E.emplId
AND De.devicePurchDate >= '2021-02-16';
```
- B

```
SELECT E.IName, E.hireDate, D.dName, De.deviceMake, De.devicePurchDate
FROM Dept D
JOIN Emp E ON D.emplId = E.emplId
JOIN Device De ON De.emplId = E.emplId
AND De.devicePurchDate >= '2021-02-16';
```
- C

```
SELECT E.IName, E.hireDate, D.dName, De.deviceMake, De.devicePurchDate
FROM Dept D
JOIN Emp E ON D.deptNo = E.emplId
JOIN Device De ON De.deviceNo = E.emplId
AND De.devicePurchDate >= '2021-02-16';
```
- D

```
SELECT E.IName, E.hireDate, D.dName, De.deviceMake, De.devicePurchDate
FROM Dept D
JOIN Emp E ON D.deviceNo = E.deptNo
JOIN Device De ON De.emplId = E.deviceNo
AND De.devicePurchDate >= '2021-02-16';
```

Question 5

Based on the Mock Logical ERD - Retrieve a list of employee numbers and surnames, departments, device makes and models for those staff who manage other employees, alongside the employee numbers and the surnames of these staff whom they actually manage.

A

```
SELECT M.empId, M.IName, D.dName, DM.deviceMake, DM.deviceModel, E.empId, E.IName
FROM Dept D
JOIN Emp M ON D.deptNo = M.mgrId
JOIN Emp E ON M.mgrId = E.empId
JOIN Device DM ON M.mgrId = DM.deviceNo
ORDER BY M.empId, D.dName, DM.deviceMake, DM.deviceModel;
```

B

```
SELECT M.empId, M.IName, D.dName, DM.deviceMake, DM.deviceModel, E.empId, E.IName
FROM Dept D
JOIN Emp M ON D.deptNo = E.empId
JOIN Emp E ON M.MgrId = E.empId
JOIN Device DM ON M.empId = DM.mgrId
ORDER BY M.empId, D.dName, DM.deviceMake, DM.deviceModel;
```

C

```
SELECT M.empId, M.IName, D.dName, DM.deviceMake, DM.deviceModel, E.empId, E.IName
FROM Dept D
JOIN Emp M ON D.deptNo = M.deptNo
JOIN Emp E ON M.empId = E.mgrId
JOIN Device DM ON M.empId = DM.empId
ORDER BY M.empId, D.dName, DM.deviceMake, DM.deviceModel;
```

D

```
SELECT M.empId, M.IName, D.dName, DM.deviceMake, DM.deviceModel, E.empId, E.IName
FROM Dept D
JOIN Emp M ON D.mgrId = M.mgrId
JOIN Emp E ON M.empId = E.empId
JOIN Device DM ON M.deviceNo = DM.deviceNo
ORDER BY M.empId, D.dName, DM.deviceMake, DM.deviceModel;
```

Question 6

Based on the Mock Logical ERD - Retrieve a list of employees who earn more than their colleagues in departments 10, 30 and 40.

A SELECT empId, lName, salary
FROM Emp
WHERE salary >
(SELECT salary
FROM Emp
WHERE deptNo = 10 OR 30 OR 40));

B SELECT empId, lName, salary
FROM Emp
WHERE salary >
(SELECT salary
FROM Emp
WHERE deptNo IN (10, 30, 40));

C SELECT empId, lName, salary
FROM Emp
WHERE salary > ALL
(SELECT salary
FROM Emp
WHERE deptNo IN (10, 30, 40));

D SELECT empId, lName, salary
FROM Emp
WHERE salary > ANY
(SELECT salary
FROM Emp
WHERE deptNo IN (10, 30, 40));

Question 7

Based on the Mock Logical ERD - Retrieve a list of employees and for each employee, the actual number of devices they have been allocated. Only include the employees who have been given at least 2 devices. Label the columns properly.

A SELECT empld AS "Emp ID", COUNT(empld) AS "Nb of Devices Owned"
FROM Device
GROUP BY (empld)
HAVING COUNT(empld) >=2;

B SELECT empld AS "Emp ID", COUNT(empld) AS "Nb of Devices Owned"
FROM Emp
GROUP BY (empld)
HAVING COUNT(empld) >=2;

C SELECT empld AS "Emp ID", deviceNo AS "Device No",
COUNT(empld) AS "Nb of Devices Owned"
FROM Device
GROUP BY (empld)
HAVING COUNT(empld) >=2;

D SELECT empld AS "Emp ID", COUNT(empld) AS "Nb of Devices Owned"
FROM Device
GROUP BY (empld)
WHERE COUNT(empld) >=2;

Question 8

Based on the Mock Logical ERD - Retrieve a list of device makes, models and prices to which a discount of 20% has been applied. Only retrieve the devices that are not faulty. Label the columns properly.

- A

SELECT deviceMake as "Make", deviceModel as "Model",
devicePrice - 0.2 * devicePrice AS "Discounted Price"
FROM Device
WHERE deviceFaults IS NULL;
- B

SELECT deviceMake as "Make", deviceModel as "Model",
devicePrice - 0.2 * devicePrice AS "Discounted Price"
FROM Device
WHERE deviceFaults = 'NULL';
- C

SELECT deviceMake as "Make", deviceModel as "Model",
(devicePrice - 0.2) * devicePrice AS "Discounted Price"
FROM Device
WHERE deviceFaults <> NULL;
- D

SELECT deviceMake as "Make", deviceModel as "Model",
(devicePrice - 0.2 * devicePrice) AS "Discounted Price"
FROM Device
WHERE deviceFaults IS NOT NULL;

Question 9

Based on the Mock Logical ERD - Retrieve a list of surnames, salaries, commissions, positions and salaries for the employees who earn more £4000 or more with a commission percentage less than 18% as well as for the employees who were hired after 24 March 2021 as programmers. Please fill in the blanks.

```
SELECT IName, salary, commPct, hireDate, position
FROM Emp
WHERE
(salary >= 4000 AND commPct < 0.18)
OR
(hireDate > '2021-03-04' AND position LIKE '%programmer%');
```

Question 10

#Based on the Mock Logical ERD - Retrieve a list of device numbers, makes and models for these devices that cost more than device number 758. Please fill in the blanks.

```
SELECT deviceNo, deviceMake, deviceModel
FROM Device
WHERE devicePrice >
  (SELECT devicePrice
   FROM Device
   WHERE deviceNo = 758)];
```

Question 11

Based on the XML document uni_marks.xml and using XPath - Return the second mark for the students who do not have "Romi" as their first name.

- ☒ **A** // student [fname != "Romi"] / mark2
- ☐ **B** // student [not(fname != "Romi")] / mark2 / fname
- ☐ **C** // student [fname = "Romi"] / name / mark2
- ☐ **D** // student [fname != "Romi"] / fname / mark2

Question 12

Based on the XML document uni_marks.xml and using XQuery - Return a list of students and their marks on the MOD101 module for the students who have scored more than 60 in the first component and less than 80 in the second component.

A

```
let $doc := doc("uni_marks.xml")
let $modcode := "MOD101"
let $firstmark := 60
let $secondmark := 80
for $S in $doc//student
where ($S//moduleCode = $modcode
and ($S/@mark2 >= $firstmark and $S/@mark1 <= $secondmark))
return $S
```

B

```
let $doc := doc("uni_marks.xml")
let $modcode := "MOD101"
let $firstmark := 60
let $secondmark := 80
for $S in $doc//student
where ($S//@moduleCode = $modcode
and ($S//mark1 >= $firstmark and $S//mark2 <= $secondmark))
return $S
```

C

```
let $doc := doc("uni_marks.xml")
let $modcode := "MOD101"
let $firstmark := 60
let $secondmark := 80
for $S in $doc//student
where ($S//@moduleCode = $modcode
and ($S/mark1 >= $firstmark and $S/mark2 <= $secondmark))
return $S
```

D

```
let $doc := doc("uni_marks.xml")
let $modcode := "MOD101"
let $firstmark := 60
let $secondmark := 80
for $S in $doc//student
where ($S//@moduleCode = $modcode
or ($S/mark1 >= $firstmark and $S/mark2 <= $secondmark))
return $S
```

Question 13

Based on the Mock Logical ERD - Retrieve a list of device numbers, makes and models for the devices that cost less than £920 and more than £1000 and for which the device make is "Pear". Customise your output so that it is displayed in one column labelled "DEVICE INFO".

A `SELECT CONCAT ('Device: ', deviceNo, ' - Make: ',deviceMake, ' - Model: ',deviceModel) AS "DEVICE INFO"`
`FROM Device`
`WHERE deviceMake LIKE '%Pear%'`
`AND (devicePrice < 920 OR devicePrice > 1000);`

B `SELECT CONCAT ('Device: ', deviceNo, ' - Make: ',deviceMake, ' - Model: ',deviceModel) AS "DEVICE INFO"`
`FROM Device`
`WHERE deviceMake LIKE '%Pear%'`
`AND (devicePrice NOT BETWEEN 920 OR 1000);`

C `SELECT CONCAT ('Device: ', deviceNo, ' - Make: ',deviceMake, ' - Model: ',deviceModel) AS "DEVICE INFO"`
`FROM Device`
`WHERE deviceMake LIKE '%Pear%'`
`AND (devicePrice < 920 AND devicePrice > 1000);`

D `SELECT CONCAT ('Device: ', deviceNo), (' - Make: ',deviceMake), (' - Model: ',deviceModel) AS "DEVICE INFO"`
`FROM Device`
`WHERE deviceMake LIKE '%Pear%'`
`AND (devicePrice NOT BETWEEN 920 AND 1000);`

Question 14

Based on the Mock Logical ERD - Retrieve a list of surnames, salaries, emails and department numbers for all employees in department 20 as well as a list of surnames, salaries, hiredates and department numbers for all employees who have a Westminster email address.

A SELECT IName, salary, email, deptNo
FROM Emp
WHERE deptNo = 20
OR email LIKE '%westminster';

B SELECT IName, salary, email, deptNo
FROM Emp
WHERE deptNo = 20
OR email LIKE '%westminster%';

C SELECT IName, salary, email, deptNo
FROM Emp
WHERE deptNo = 20
AND email LIKE '%westminster%';

D SELECT IName, salary, email, deptNo
FROM Emp
WHERE deptNo = 20
AND email = 'westminster';

Question 15

Based on the Mock Logical ERD - Display a list of surnames, emails and positions for those employees who have managers, alongside the surnames, emails and positions of the staff who actually manage them.

A SELECT Em.IName, Em.email, Em.Position,Mg.IName, Mg.email, Mg.Position
FROM Emp Mg JOIN Emp Em
ON Mg.empld = Em.empld;

B SELECT Em.IName, Em.email, Em.Position,Mg.IName, Mg.email, Mg.Position
FROM Emp Mg JOIN Emp Em
ON Mg.mgrld = Em.empld;

C SELECT Em.IName, Em.email, Em.Position,Mg.IName, Mg.email, Mg.Position
FROM Emp Mg JOIN Emp Em
ON Mg.mgrld = Em.mgrld;

D SELECT Em.IName, Em.email, Em.Position,Mg.IName, Mg.email, Mg.Position
FROM Emp Mg JOIN Emp Em
ON Mg.empld = Em.mgrld;

Question 16

Based on the Mock Logical ERD - Retrieve first names, surnames and positions of employees who earn more than £4500 and whose first name starts with the letter T.

A SELECT fName, lName, position
FROM Emp
WHERE salary > 4500 AND fName LIKE '%T';

B SELECT fName, lName, position
FROM Emp
WHERE salary < 4500 OR fName LIKE '%T';

C SELECT fName, lName, position
FROM Emp
WHERE salary > 4500 AND fName LIKE 'T%';

D SELECT *
FROM Emp
WHERE salary > 4500 AND fName LIKE '_T%';

Question 17

Based on the Mock Logical ERD - Retrieve a list of surnames, salaries, and device makes for those staff who earn less than £5200 and have been allocated a device of make 'Pear'.

```
SELECT E.IName, E.salary, D.deviceMake  
FROM Emp E JOIN Device D  
ON D.empld = D.deviceNo  
AND E.salary < 5200  
AND D.deviceMake LIKE '%Pear%';
```

```
SELECT E.IName, E.salary, D.deviceMake  
FROM Emp E JOIN Device D  
ON E.empld = D.empld  
AND (E.salary < 5200  
OR D.deviceMake LIKE '%Pear%');
```

```
SELECT E.IName, E.salary, D.deviceMake  
FROM Emp E JOIN Device D  
ON E.deviceNo = D.deviceNo  
AND E.salary < 5200  
AND D.deviceMake LIKE '%Pear%';
```

```
SELECT E.IName, E.salary, D.deviceMake  
FROM Emp E JOIN Device D  
ON E.empld = D.empld  
AND E.salary < 5200  
AND D.deviceMake LIKE '%Pear%';
```


Question 18

Based on the Mock Logical ERD - Retrieve a list of department names, locations and creation dates but only for the departments created in 2021 and for these that have not been created yet.

A SELECT dName, dLoc, createDate
FROM Dept
WHERE createDate LIKE '%2021%'
AND createDate IS NULL;

B SELECT dName, dLoc, createDate
FROM Dept
WHERE createDate BETWEEN '2021-01-01' AND '2021-12-31'
OR createDate IS NULL;

C SELECT dName, dLoc, createDate
FROM Dept
WHERE (createDate >= '2021-01-01' OR createDate <= '2021-12-31')
AND createDate IS NULL;

D SELECT dName, dLoc, createDate
FROM Dept
WHERE createDate >= '2021-01-01' AND <= '2021-12-31'
OR createDate IS NULL;

Question 19

Based on the Mock Logical ERD - Retrieve a list of employee numbers, surnames, emails and department numbers for the employees who have emails that finish in "global.com" and "local.com" and also who work in department 20 or are managed by employee 102. Please fill in the blanks.

```
SELECT empId, lName, email, deptNo
FROM Emp
WHERE
(email LIKE '%global.com' OR email LIKE '%local.com')
AND
(deptNo = 20 OR mgrId = 102);
```

Question 20

Based on the XML document uni_marks.xml and using XPath - Return the names of the modules led by Francky Bandini.

- A** // module [moduleleader = "Francky Bandini"] / modulename
- B // module [moduleleader = "Francky Bandini"] / module / modulename
- C // module [moduleleader = "Francky Bandini"] / modulename / moduleleader
- D // module [moduleleader = "Francky Bandini"] / module / name

Question 21

#Based on the Mock Logical ERD - Retrieve a list of device makes (or brands) and for each device make, the earliest date and the latest date at which the devices of that make were purchased. Please fill in the blanks.

```
SELECT deviceMake, MIN(devicePurchDate), MAX(devicePurchDate)  
FROM Device  
GROUP BY deviceMake;
```

Question 22

Based on the Mock Logical ERD - Retrieve a list of employee first names, positions and email with the names and locations of the departments in which they work. Also add the departments that do not have any employees working in them.

A SELECT E.fName, E.position, E.email, D.dName, dLoc
FROM Emp E LEFT OUTER JOIN Dept D
ON E.empNo = D.deptNo;

B SELECT E.fName, E.position, E.email, D.dName, dLoc
FROM Emp E RIGHT OUTER JOIN Dept D
ON E.deptNo = D.deptNo;

C SELECT E.fName, E.position, E.email, D.dName, dLoc
FROM Emp E RIGHT OUTER JOIN Dept D
ON D.empNo = D.empNo;

D SELECT E.fName, E.position, E.email, D.dName, dLoc
FROM Emp E LEFT OUTER JOIN Dept D
ON E.deptNo = D.deptNo;

Question 23

Based on the XML document uni_marks.xml and using XPath - Return all the students who have scored more than 43 in the coursework on the Java Programming module.

- ☐ **A** // module [modulename = "Java Programming" and mark1 [@type = "cwk"] and mark1 > 43]
- ☐ **B** // module [modulename = "Java Programming"] / student / mark1 [@type = "cwk"] and [mark1 > 43]
- ☒ **C** // module [modulename = "Java Programming"] / student [mark1 [@type = "cwk"] and mark1 > 43]
- ☐ **D** // module [modulename = "Java Programming"] | student [mark1 [@type = "cwk"] and mark1 > 43]

Final Mark
Last attempt with a mark

96 / 96

 Attempt 29 Submitted 17/01/2025, 02:40 (UTC+5:30)

96 / 96

 Question scores appear after all marks have been posted

Assignment Content

Question 1

Based on the Mock Logical ERD - Retrieve a list of device makes and models with the names and position of staf to whom they have been allocated. Also add the staff who have not been allocated a device.

A SELECT De.deviceMake, De.deviceModel, Em.IName, Em.position
FROM Device De LEFT OUTER JOIN Emp Em
ON Em.deviceId = De.deviceId;

 Feedback

Feedback for student
Your instructor hasn't added feedback